

Legendre's Equation & Polynomials

The differential equation,

$$(1-x^2) \frac{d^2 y}{dx^2} - 2x \frac{dy}{dx} + n(n+1)y = 0$$

where n is a real number, is called Legendre's Differential Equation and its solutions are known as Legendre's Functions or Legendre's Polynomials.

Legendre's Function of First Kind $P_n(x)$

When ' n ' is a positive integer, the ^{first} solution is denoted by $P_n(x)$ and is called Legendre's function of the first kind.

$$P_n(x) = \frac{1 \cdot 3 \cdot 5 \dots (2n-1)}{n!} \left[x^n - \frac{n(n-1)}{2(2n-1)} x^{n-2} + \dots \right]$$

$P_n(x)$ is a terminating series. Also, $P_n(1) = 1$.

Legendre's Function of the Second Kind $Q_n(x)$

When 'n' is a positive integer, the second solution is denoted by $Q_n(x)$ and is called Legendre's function of the second kind.

$$Q_n(x) = \frac{n!}{1 \cdot 3 \cdot 5 \dots (2n+1)} \left[x^{-(n+1)} + \frac{(n+1)(n+2)}{2(2n+3)} x^{-(n+3)} + \dots \right]$$

$Q_n(x)$ is a non-terminating series.

General Solution

The general solution is,

$$y = A \cdot P_n(x) + B \cdot Q_n(x).$$

where, A & B are arbitrary constants.

Generating Function for $P_n(x)$

The generating function for $P_n(x)$ is $(1-2xz+z^2)^{-1/2}$. $P_n(x)$ is the coefficient of z^n in the expansion of $(1-2xz+z^2)^{-1/2}$.

$$\text{i.e. } (1-2xz+z^2)^{-1/2} = \sum_{n=0}^{\infty} P_n(x) z^n$$

Questions

Imp.

- Q.1) Show that: (i) $P_n(1) = 1$
(ii) $P_n(-x) = (-1)^n P_n(x)$
(iii) $P'_n(-x) = (-1)^{n+1} P'_n(x)$

Solⁿ:

We know,

$$\sum_{n=0}^{\infty} P_n(x) z^n = (1-2xz+z^2)^{-1/2} \quad \text{--- (1)}$$

i. Putting $x=1$,

$$\sum_{n=0}^{\infty} P_n(1) z^n = (1-2z+z^2)^{-1/2}$$

$$= [(1-z)^2]^{-1/2}$$

$$= (1-z)^{-1}$$

$$\Rightarrow \sum_{n=0}^{\infty} P_n(1) z^n = 1 + z + z^2 + \dots + z^n + \dots$$

$$\Rightarrow \sum_{n=0}^{\infty} P_n(1) z^n = \sum_{n=0}^{\infty} 1 \cdot z^n$$

$$\Rightarrow P_n(1) = 1. \quad [\text{By equating the coefficient of } z^n]$$

ii. Putting $x = -x$,

$$\sum_{n=0}^{\infty} P_n(-x) z^n = (1 + 2xz + z^2)^{-1/2} \quad \text{--- (2)}$$

Replacing z by $-z$ in eqⁿ. (1),

$$\sum_{n=0}^{\infty} P_n(x) (-z)^n = (1 + 2xz + z^2)^{-1/2}$$

$$\Rightarrow \sum_{n=0}^{\infty} (-1)^n P_n(x) z^n = (1 + 2xz + z^2)^{-1/2} \quad \text{--- (3)}$$

From (2) & (3),

$$\sum_{n=0}^{\infty} P_n(-x) z^n = \sum_{n=0}^{\infty} (-1)^n P_n(x) z^n$$

$$\Rightarrow P_n(-x) = (-1)^n P_n(x).$$

iii. We have,
$$P_n(-x) = (-1)^n P_n(x)$$

Diff. w.r.to x ,

$$-P_n'(-x) = (-1)^n P_n'(x)$$

$$\Rightarrow P_n'(-x) = (-1)^{n+1} P_n'(x).$$

Q.2) Show that: (i) $P_{2n}(0) = (-1)^n \frac{(2n)!}{2^{2n} (n!)^2}$
(ii) $P_{2n+1}(0) = 0.$

Solⁿ:

We have,

$$\sum_{n=0}^{\infty} P_n(x) z^n = (1 - 2xz + z^2)^{-1/2}$$

Putting $x=0$,

$$\sum_{n=0}^{\infty} P_n(0) z^n = (1 + z^2)^{-1/2}$$

$$= 1 - \frac{z^2}{2} + \frac{1 \cdot 3}{2 \cdot 4} z^4 - \dots + \dots$$

$$+ (-1)^n \frac{1 \cdot 3 \dots (2n-1)}{2 \cdot 4 \dots (2n)} \cdot z^{2n}$$

i. Equating the coefficients of z^{2n} ,

$$P_{2n}(0) = (-1)^n \frac{1 \cdot 3 \cdot 5 \dots (2n-1)}{2 \cdot 4 \cdot 6 \dots (2n)}$$

$$\Rightarrow P_{2n}(0) = (-1)^n \frac{1 \cdot 2 \cdot 3 \cdot 4 \cdot 5 \dots (2n-1)(2n)}{[2 \cdot 4 \cdot 6 \dots (2n)]^2}$$

$$\Rightarrow P_{2n}(0) = (-1)^n \frac{(2n)!}{[2^n \cdot 1 \cdot 2 \cdot 3 \dots (n)]^2}$$

$$\Rightarrow P_{2n}(0) = (-1)^n \frac{(2n)!}{2^{2n} (n!)^2}$$

ii. Equating the coefficients of z^{2n+1} , we get,

$$P_{2n+1}(0) = 0$$

Since, the RHS contains only even power of z .

Q.3) Prove that:

$$(i) \sum_{n=0}^{\infty} P_n(x) = \frac{1}{\sqrt{2-2x}}$$

Solⁿ:

We have,

$$\sum_{n=0}^{\infty} P_n(x) z^n = (1 - 2xz + z^2)^{-1/2}$$

Putting $z=1$,

$$\sum_{n=0}^{\infty} P_n(x) \cdot 1 = (1 - 2x + 1)^{-1/2}$$

$$\Rightarrow \sum_{n=0}^{\infty} P_n(x) \cdot \cancel{(1-2x)} = (2-2x)^{-1/2}$$

$$\Rightarrow \sum_{n=0}^{\infty} P_n(x) = \frac{1}{\sqrt{2-2x}}$$

(ii) $P_n(-1) = (-1)^n$

Solⁿ:

We have,

$$P_n(-x) = (-1)^n P_n(x)$$

Putting $x=1$,

$$P_n(-1) = (-1)^n P_n(1)$$

$$\Rightarrow P_n(-1) = (-1)^n \cdot 1 \quad [P_n(1) = 1]$$

Q.4) Prove:

(i) $P'_n(1) = \frac{n(n+1)}{2}$

Solⁿ:

We have, $y = P_n(x)$ is the solution of,

$$(1-x^2)y'' - 2xy' + n(n+1)y = 0$$

$$\Rightarrow (1-x^2)P''_n(x) - 2xP'_n(x) + n(n+1)P_n(x) = 0$$

Putting $x=1$,

$$-2P'_n(1) + n(n+1)P_n(1) = 0$$

$$\Rightarrow P'_n(1) = \frac{n(n+1)}{2} \quad [P_n(1) = 1]$$

$$(ii) P'_n(-1) = (-1)^{n+1} \cdot \frac{n(n+1)}{2}$$

Solⁿ:

We have, $y = P_n(x)$ is the solution of

$$(1-x^2)y'' - 2xy' + n(n+1)y = 0$$

$$\Rightarrow (1-x^2)P''_n(x) - 2xP'_n(x) + n(n+1)P_n(x) = 0$$

Putting $x = -1$,

$$2P'_n(-1) = -n(n+1)P_n(-1)$$

$$\Rightarrow P'_n(-1) = \frac{-n(n+1)}{2} \times (-1)^n \quad \left[\begin{array}{l} \Rightarrow P_n(-1) = (-1)^n \\ P_n(-x) = (-1)^n P_n(x) \end{array} \right]$$

$$\Rightarrow P'_n(-1) = (-1)^{n+1} \cdot \frac{n(n+1)}{2}$$

Imp # Rodrigue's Formula

The relation below is called Rodrigue's formula:

$$P_n(x) = \frac{1}{2^n n!} \cdot \frac{d^n}{dx^n} (x^2 - 1)^n$$

Proof:

$$\text{Let } v = (x^2 - 1)^n$$

$$\Rightarrow \frac{dv}{dx} = 2nx(x^2 - 1)^{n-1}$$

$$\Rightarrow \frac{dv}{dx} = \frac{2nx(x^2 - 1)^n}{(x^2 - 1)^1}$$

$$\Rightarrow \underbrace{(x^2 - 1)}_v \underbrace{\frac{dv}{dx}}_u = 2n \underbrace{vx}_u \quad [\because v = (x^2 - 1)^n] \quad (1)$$

Differentiating above eqⁿ. (n+1) times using Leibniz's Theorem, we get,

$$\frac{(x^2 - 1) d^{n+2} v}{dx^{n+2}} + (n+1) (2x) \frac{d^{n+1} v}{dx^{n+1}} + \frac{(n+1)n (2)}{2!} \frac{d^n v}{dx^n}$$

$$= 2n \left[x \frac{d^{n+1} v}{dx^{n+1}} + (n+1) \frac{d^n v}{dx^n} \right]$$

Leibnitz Theorem:

$$\frac{d^n}{dx^n} (uv) = u_n v + n u_{n-1} v_1 + \frac{n(n-1)}{2} u_{n-2} v_2 + \dots \quad {}^nC_n uv_n$$

~~(1-x)~~

$$(x^2-1) \frac{d^{n+2} v}{dx^{n+2}} + 2nx \frac{d^{n+1} v}{dx^{n+1}} + 2x \frac{d^{n+1} v}{dx^{n+1}} + n(n+1) \frac{d^n v}{dx^n}$$

$$= 2nx \frac{d^{n+1} v}{dx^{n+1}} + 2n(n+1) \frac{d^n v}{dx^n}$$

$$\Rightarrow (x^2-1) \frac{d^{n+2} v}{dx^{n+2}} + 2x \frac{d^{n+1} v}{dx^{n+1}} - n(n+1) \frac{d^n v}{dx^n} = 0$$

$$\Rightarrow (1-x^2) \frac{d^{n+2} v}{dx^{n+2}} - 2x \frac{d^{n+1} v}{dx^{n+1}} + n(n+1) \frac{d^n v}{dx^n} = 0$$

The above equation is in the form of Legendre's Differential Equation - and $\frac{d^n v}{dx^n}$ is its solution.

We know, the solution of Legendre's equation is $P_n(x)$. So,

$$\frac{d^n v}{dx^n} = C \cdot P_n(x) \quad \text{--- (2)}$$

where, C is constant.

We have,

$$v = (x^2 - 1)^n$$

$$\Rightarrow v = (x^2 - 1)^n (x + 1)^n \quad \left\{ (x^2 - 1)^n = [(x^2 - 1)(x + 1)]^n \right\}$$

Differentiating 'n' times

$$\begin{aligned} \frac{d^n v}{dx^n} &= (x+1)^n \frac{d^n (x-1)^n}{dx^n} + n \cdot n (x+1)^{n-1} \frac{d^{n-1} (x-1)^n}{dx^{n-1}} \\ &\quad + \dots + (x-1)^n \frac{d^n (x+1)^n}{dx^n} \quad \text{--- (3)} \end{aligned}$$

Putting $x=1$ in eqⁿ. (2),

$$\left(\frac{d^n v}{dx^n} \right)_{x=1} = C P_n(1) = C \quad [\because P_n(1) = 1] \quad \text{--- (4)}$$

Putting $x=1$ in eqⁿ. (3),

$$\left(\frac{d^n v}{dx^n} \right)_{x=1} = 2^n n! + 0 + \dots + 0 = 2^n n! \quad \text{--- (5)}$$

From (4) & (5),

$$C = 2^n n!$$

Putting this value of C in eqⁿ. (2), we get,

$$P_n(x) = \frac{1}{2^n n!} \cdot \frac{d^n}{dx^n} (x^2 - 1)^n \quad \underline{\text{proved}}$$

Putting $n = 0, 1, 2, 3, \dots$ in above formula, we get Legendre's Polynomials,

$$P_0(x) = 1$$

$$P_1(x) = \frac{1}{2} \frac{d(x^2 - 1)}{dx} = x$$

$$P_2(x) = \frac{1}{2} (3x^2 - 1)$$

$$P_3(x) = \frac{1}{2} (5x^3 - 3x)$$

$\vdots$

And so on...

Questions

Q.1) Show that: $\int_{-1}^1 P_n(x) dx = 0$, $n \neq 0$

$$\& \int_{-1}^1 P_n(x) dx = 2, \quad n = 0$$

Solⁿ:

We have,

$$P_n(x) = \frac{1}{2^n n!} \frac{d^n}{dx^n} (x^2 - 1)^n$$

$$\int_{-1}^1 P_n(x) dx = \int_{-1}^1 \frac{1}{2^n n!} \frac{d^n}{dx^n} (x^2 - 1)^n dx \quad (1)$$

$$\Rightarrow \int_{-1}^1 P_n(x) dx = \frac{1}{2^n n!} \left[\frac{d^{n-1}}{dx^{n-1}} (x^2 - 1)^n \right]_{-1}^1$$

$$\Rightarrow \int_{-1}^1 P_n(x) dx = \frac{1}{2^n n!} [0 - 0]$$

$$\Rightarrow \int_{-1}^1 P_n(x) dx = 0 \quad \underline{\underline{\text{proved}}}$$

Putting $n=0$ in eqⁿ. (1),

$$\int_{-1}^1 P_0(x) dx = \frac{1}{2^0 0!} \int_{-1}^1 \frac{d^0}{dx^0} (x^2-1)^0 dx$$

$$\Rightarrow \int_{-1}^1 P_0(x) dx = \int_{-1}^1 1 \cdot dx$$

$$\Rightarrow \int_{-1}^1 P_0(x) dx = [x]_{-1}^1 = 1 - (-1)$$

$$\Rightarrow \int_{-1}^1 P_0(x) dx = 2 \quad \underline{\text{proved}}$$

Q.2) Evaluate: $\int_{-1}^1 x^2 P_2(x) dx$

Solⁿ:

$$\int_{-1}^1 x^2 P_2(x) dx$$

$$= \int_{-1}^1 x^2 \cdot \frac{1}{2} (3x^2-1) dx$$

$$= \frac{1}{2} \int_{-1}^1 (3x^4 - x^2) dx$$

$$= \frac{1}{2} \left[\frac{3x^5}{5} - \frac{x^3}{3} \right]_{-1}^1$$

$$= \frac{1}{2} \left[\frac{3}{5} - \frac{1}{3} - \left(-\frac{3}{5} + \frac{1}{3} \right) \right]$$

$$= \frac{1}{2} \left[\frac{3}{5} - \frac{1}{3} + \frac{3}{5} - \frac{1}{3} \right]$$

$$= \frac{1}{2} \left[\frac{6}{5} - \frac{2}{3} \right]$$

$$= \frac{1}{2} \times \frac{18-10}{15}$$

$$= \frac{8}{2 \times 15}$$

$$= \frac{4}{15}$$

Q.3) Express the polynomial $f(x) = 4x^3 - 2x^2 - 3x + 8$ in terms of Legendre Polynomials.

Solⁿ:

$$\text{Let, } 4x^3 - 2x^2 - 3x + 8 = a P_3(x) + b P_2(x) + c P_1(x) + d P_0(x) \quad (1)$$

$$\Rightarrow 4x^3 - 2x^2 - 3x + 8 = a \left(\frac{5x^3 - 3x}{2} \right) + b \left(\frac{3x^2 - 1}{2} \right) + cx + d$$

$$= \frac{5ax^3}{2} - \frac{3ax}{2} + \frac{3bx^2}{2} - \frac{b}{2} + cx + d$$

Equating the like terms,

$$\frac{5a}{2} = 4 \Rightarrow a = \frac{8}{5}$$

$$\frac{3b}{2} = -2 \Rightarrow b = -\frac{4}{3}$$

$$-\frac{3a}{2} + c = -3 \Rightarrow c = -3 + \frac{3 \times 8}{2 \times 5} = -\frac{3}{5}$$

$$-\frac{b}{2} + d = 8 \Rightarrow d = 8 + \frac{(-4/3)}{2} = 8 - \frac{2}{3} = \frac{22}{3}$$

∴ Eqⁿ (1) becomes,

$$4x^3 - 2x^2 - 3x + 8 = \frac{8}{5} P_3(x) - \frac{4}{3} P_2(x) - \frac{3}{5} P_1(x) + \frac{22}{3} P_0(x).$$

Q.4) Express $f(x) = x^4 + 3x^3 - x^2 + 5x - 2$ in terms of Legendre Polynomials.

Solⁿ:

$$\begin{aligned} \text{Let, } x^4 + 3x^3 - x^2 + 5x - 2 \\ = a P_4(x) + b P_3(x) + c P_2(x) + d P_1(x) + e P_0(x) \end{aligned} \quad (1)$$

$$= a \left(\frac{35x^4 - 30x^2 + 3}{8} \right) + b \left(\frac{5x^3 - 3x}{2} \right) + c \left(\frac{3x^2 - 1}{2} \right) + dx + e$$

$$= \frac{35a}{8} x^4 - \frac{30a}{8} x^2 + \frac{3a}{8} + \frac{5b}{2} x^3 - \frac{3b}{2} x$$

$$+ \frac{3c}{2} x^2 - \frac{c}{2} + dx + e$$

$$\begin{aligned} = \frac{35a}{8} x^4 + \frac{5b}{2} x^3 - \left(\frac{30a}{8} - \frac{3c}{2} \right) x^2 \\ - \left(\frac{3b}{2} - d \right) x + \left(\frac{3a}{8} - \frac{c}{2} + e \right) \end{aligned}$$

-Equating the like terms,

$$\textcircled{1} \quad \frac{35a}{8} = 1 \Rightarrow a = \frac{8}{35}$$

$$\textcircled{2} \quad \frac{5b}{2} = 3 \Rightarrow b = \frac{6}{5}$$

$$\textcircled{3} \quad -\frac{30a}{8} + \frac{3c}{2} = -1$$

$$\Rightarrow \frac{3c}{2} = -1 + \frac{30}{8} \left(\frac{8}{35} \right) = -1 + \frac{6}{7}$$

$$\Rightarrow \frac{3c}{2} = \frac{-1}{7}$$

$$\Rightarrow c = \frac{-2}{21}$$

$$\textcircled{4} \quad -\frac{3b}{2} + d = 5$$

$$\Rightarrow d = 5 + \frac{3b}{2} = 5 + \frac{3}{2} \left(\frac{6}{5} \right) = 5 + \frac{9}{5}$$

$$\Rightarrow d = \frac{34}{5}$$

$$\textcircled{5} \quad \frac{3a}{8} - \frac{c}{2} + c = -2$$

$$\Rightarrow c = -2 - \frac{3a}{8} + \frac{c}{2}$$

$$\Rightarrow c = -2 - \frac{3}{8} \left(\frac{8}{35} \right) + \frac{1}{2} \left(\frac{-2}{21} \right)$$

$$\Rightarrow c = -2 - \frac{3}{35} - \frac{1}{21}$$

$$\Rightarrow c = -\frac{73}{35} - \frac{1}{21}$$

$$\Rightarrow c = \frac{-1533 - 35}{735} = \frac{-1568}{735} = \frac{-32}{15}$$

~~$$\Rightarrow c = \frac{-1533 - 35}{735}$$~~

$\therefore$ Egⁿ. (1) becomes,

$$\frac{8}{35} P_4(x) + \frac{6}{5} P_3(x) - \frac{2}{21} P_2(x) + \frac{34}{5} P_1(x) - \frac{32}{15} P_0(x)$$

Imp.
#

Recurrence Relations

$$I \quad (n+1) P_{n+1}(x) = (2n+1)x P_n(x) - n P_{n-1}(x)$$

Proof:

We have,

$$\sum_{n=0}^{\infty} P_n(x) z^n = (1-2xz+z^2)^{-1/2} \quad \text{--- (1)}$$

Differentiating above eqⁿ. w.r.to z ,

$$\sum_{n=0}^{\infty} n z^{n-1} P_n(x) = \frac{-1}{2} (1-2xz+z^2)^{-3/2} \cdot (-2x+2z)$$

$$\Rightarrow \sum_{n=0}^{\infty} n P_n(x) z^{n-1} = \frac{-1}{2} (1-2xz+z^2)^{-3/2} \cdot (-2(x-z))$$

$$\Rightarrow \sum_{n=0}^{\infty} n P_n(x) z^{n-1} = (1-2xz+z^2)^{-3/2} \cdot (x-z)$$

Multiplying both sides by $(1-2xz+z^2)$,

$$(1-2xz+z^2) \sum_{n=0}^{\infty} n P_n(x) z^{n-1} = (1-2xz+z^2)^{-1/2} \cdot (x-z)$$

$$\Rightarrow (1-2xz+z^2) \sum_{n=0}^{\infty} n P_n(x) z^{n-1} = (x-z) \sum_{n=0}^{\infty} P_n(x) z^n$$

[From eqⁿ. (1)]

$$\Rightarrow \sum n P_n(x) z^{n-1} - \sum 2nx P_n(x) z^n$$

$$+ \sum n P_n(x) z^{n+1} = \sum x P_n(x) z^n - \sum P_n(x) z^{n+1}$$

→ Making all z^n by replacing 'n',

$$\sum (n+1) P_{n+1}(x) z^n - \sum 2nx P_n(x) z^n + \sum (n-1) P_{n-1}(x) z^n$$

$$= \sum x P_n(x) z^n - \sum P_{n-1}(x) z^n$$

Equating the coefficients of z^n , we get,

$$(n+1) P_{n+1}(x) - 2nx P_n(x) + (n-1) P_{n-1}(x)$$

$$= x P_n(x) - P_{n-1}(x)$$

$$\Rightarrow (n+1) P_{n+1}(x) = (2n+1)x P_n(x) - n P_{n-1}(x).$$

II $n P_n(x) = x P_n'(x) - P_{n-1}'(x)$

Proof:

We have,

$$\sum_{n=0}^{\infty} P_n(x) z^n = (1 - 2xz + z^2)^{-1/2} \quad \text{--- (1)}$$

Diff. eqⁿ. (1) w.r.to z ,

$$\sum n P_n(x) z^{n-1} = \frac{-1}{2} (1 - 2xz + z^2)^{-3/2} \cdot (-2x + 2z)$$

$$\Rightarrow \sum n P_n(x) z^{n-1} = (x - z) \cdot (1 - 2xz + z^2)^{-3/2} \quad \text{--- (2)}$$

Diff. eqⁿ. (1) w.r.to x ,

$$\sum P_n'(x) z^n = \frac{-1}{2} (1 - 2xz + z^2)^{-3/2} \cdot (-2z)$$

$$\Rightarrow \sum P_n'(x) z^n = z (1 - 2xz + z^2)^{-3/2} \quad \text{--- (3)}$$

Dividing eqⁿ. (2) by eqⁿ. (3), we get,

$$\frac{\sum n P_n(x) z^{n-1}}{\sum P_n'(x) z^n} = \frac{x - z}{z}$$

$$\Rightarrow \sum n P_n(x) z^n = \sum x P_n'(x) z^n - \sum P_n'(x) z^{n+1}$$

$$\Rightarrow \sum n P_n(x) z^n = \sum x P_n'(x) z^n - \sum P_{n-1}'(x) z^n$$

Equating the coefficients of z^n , we get,

$$n P_n(x) = x P_n'(x) - P_{n-1}'(x).$$

III. $(2n+1) P_n(x) = P_{n+1}'(x) - P_{n-1}'(x)$

Proof:

From Relation (I),

$$(n+1) P_{n+1}(x) = (2n+1) x P_n(x) - n P_{n-1}(x) \quad \text{--- (1)}$$

Diff. w.r. to x ,

$$(n+1) P_{n+1}'(x) = (2n+1) P_n(x) + (2n+1) x P_n'(x) - n P_{n-1}'(x) \quad \text{--- (2)}$$

From Relation (II),

$$n P_n(x) = x P_n'(x) - P_{n-1}'(x)$$

$$\Rightarrow x P_n'(x) = n P_n(x) + P_{n-1}'(x) \quad \text{--- (3)}$$

Putting value of $x P_n'(x)$ into eqⁿ. (2),

$$(n+1) P_{n+1}'(x) = (2n+1) P_n(x) + (2n+1) [n P_n(x) + P_{n-1}'(x)] - n P_{n-1}'(x)$$

$$\Rightarrow (n+1) P'_{n+1}(x) = (2n+1) P_n(x) + n(2n+1) P_n(x) + (2n+1) P'_{n-1}(x) - n P'_{n-1}(x)$$

$$\Rightarrow (n+1) P'_{n+1}(x) = (2n+1)(n+1) P_n(x) + (n+1) P'_{n-1}(x)$$

$$\Rightarrow P'_{n+1}(x) = (2n+1) P_n(x) + P'_{n-1}(x)$$

$$\Rightarrow (2n+1) P_n(x) = P'_{n+1}(x) - P'_{n-1}(x).$$

IV. $P'_n(x) = x P'_{n-1}(x) + n P_{n-1}(x)$

Proof:

From Relation (I),

$$(n+1) P_{n+1}(x) = (2n+1) x P_n - n P_{n-1}(x) \quad \text{--- (1)}$$

Replacing n by $(n-1)$, we get,

$$n P_n = (2n-1) x P_{n-1} - (n-1) P_{n-2}$$

Diff. w.r.to x ,

$$n P'_n = (2n-1) P_{n-1} + (2n-1) x P'_{n-1} - (n-1) P'_{n-2}$$

~~$$n P'_n = (2n-1) P_{n-1} + (2n-1) x P'_{n-1} - (n-1) P'_{n-2}$$~~

$$\Rightarrow n P'_n = (2n-1) P_{n-1} + n x P'_{n-1} + (n-1) x P'_{n-1} - (n-1) P'_{n-2}$$

$$\Rightarrow n [P'_n - x P'_{n-1}] = (2n-1) P_{n-1} + (n-1) [x P'_{n-1} - P'_{n-2}] \quad (2)$$

From Relation (II),

$$x P'_n(x) - P'_{n-1}(x) = n P_n(x)$$

Replacing n by $(n-1)$,

$$x P'_{n-1} - P'_{n-2} = (n-1) P_{n-1} \quad (3)$$

From (2) & (3)

$$n [P'_n - x P'_{n-1}] = (2n-1) P_{n-1} + (n-1)(n-1) P_{n-1}$$

$$\Rightarrow n [P'_n - x P'_{n-1}] = [2n-1 + (n-1)(n-1)] P_{n-1}$$

$$\Rightarrow n [P'_n - x P'_{n-1}] = [2n-1 + n^2 - 2n + 1] P_{n-1}$$

$$\Rightarrow n [P'_n - x P'_{n-1}] = n^2 P_{n-1}$$

$$\Rightarrow P'_n(x) = x P'_{n-1}(x) + n P_{n-1}(x).$$

V. $(x^2-1) P_n'(x) = n[x P_n(x) - P_{n-1}(x)]$

Proof:

Relation II,

$$\begin{aligned} x P_n'(x) - P_{n-1}'(x) &= n P_n(x) \\ \Rightarrow P_{n-1}'(x) &= x P_n'(x) - n P_n(x) \quad \text{--- (1)} \end{aligned}$$

Relation IV,

$$\begin{aligned} P_n'(x) - x P_{n-1}'(x) &= n P_{n-1}(x) \\ \Rightarrow P_n'(x) - x [x P_n'(x) - n P_n(x)] &= n P_{n-1}(x) \\ \Rightarrow P_n'(x) - x^2 P_n'(x) + nx P_n(x) &= n P_{n-1}(x) \\ \Rightarrow (1-x^2) P_n'(x) &= n P_{n-1}(x) - nx P_n(x) \\ \Rightarrow -(x^2-1) P_n'(x) &= n [P_{n-1}(x) - x P_n(x)] \\ \Rightarrow (x^2-1) P_n'(x) &= n [x P_n(x) - P_{n-1}(x)] . \end{aligned}$$

$$\text{VI. } (x^2-1) P_n'(x) = (n+1) [P_{n+1}(x) - x P_n(x)]$$

Proof:

Relation I,

$$(n+1) P_{n+1}(x) = (2n+1)x P_n(x) - n P_{n-1}(x)$$

$$\Rightarrow (n+1) [P_{n+1} - x P_n] = n (x P_n - P_{n-1}) \quad \text{--- (1)}$$

$$[(2n+1) - n] = (n+1)$$

Relation V,

$$(x^2-1) P_n'(x) = \cancel{(n+1)} n [x P_n(x) - P_{n-1}(x)] \quad \text{--- (2)}$$

From eqⁿ. (1) & (2),

$$(x^2-1) P_n'(x) = (n+1) [P_{n+1}(x) - x P_n(x)].$$

Imp. # Orthogonality of Legendre Polynomials

i. $\int_{-1}^1 P_m(x) P_n(x) dx = 0, m \neq n$

Proof:

We know, $y = P_n(x)$ & $z = P_m(x)$ are the solutions of the equations

$$(1-x^2)y'' - 2xy' + n(n+1)y = 0 \quad \text{--- (1)}$$

$$\& (1-x^2)z'' - 2xz' + m(m+1)z = 0 \quad \text{--- (2)}$$

Now, $z \times (1) - y \times (2),$

$$(1-x^2)[zy'' - yz''] - 2x[zy' - yz'] + [n(n+1) - m(m+1)]yz = 0$$

$$\Rightarrow \frac{d}{dx} [(1-x^2)(zy' - yz')] + [n^2 + n - m^2 - m]yz = 0$$

$$\left[\because \frac{d}{dx} [(1-x^2)(zy' - yz')] = (1-x^2)(zy'' + z'y' - yz'' - y'z') - 2x(zy' - yz') \right. \\ \left. = (1-x^2)(zy'' - yz'') - 2x(zy' - yz') \right]$$

$$\frac{d}{dx} [(1-x^2)(zy' - yz')] + [(n+m)(n-m) + n-m]yz = 0$$

$$\Rightarrow \frac{d}{dx} [(1-x^2)(zy' - yz')] + (n-m)(n+m+1)yz = 0$$

$$\Rightarrow \int_{-1}^1 \frac{d}{dx} [(1-x^2)(zy' - yz')] + (n-m)(n+m+1) \int_{-1}^1 yz \, dx = 0$$

$$\Rightarrow [(1-x^2)(zy' - yz')]_{-1}^1 + (n-m)(n+m+1) \int_{-1}^1 yz \, dx = 0$$

$$\Rightarrow [0-0] + (n-m)(n+m+1) \int_{-1}^1 P_n(x) \cdot P_m(x) \, dx = 0$$

• Since, $n \neq m \Rightarrow (n-m)(n+m+1) \neq 0$

$$\int_{-1}^1 P_n(x) \cdot P_m(x) \, dx = 0 \quad \underline{\text{proved}}$$

ii. $\int_{-1}^1 [P_n(x)]^2 dx = \frac{2}{2n+1}$

Proof:

We have,

$$\sum_{n=0}^{\infty} P_n(x) z^n = (1 - 2xz + z^2)^{-1/2}$$

Squaring both sides,

$$\left(\sum_{n=0}^{\infty} P_n(x) z^n \right)^2 = \left[(1 - 2xz + z^2)^{-1/2} \right]^2$$

$$\Rightarrow \sum (P_n(x))^2 z^{2n} - 2 \sum P_n(x) P_m(x) z^{m+n} = (1 - 2xz + z^2)^{-1}$$

Integrating both sides with limits -1 to 1 ,

$$\sum z^{2n} \int_{-1}^1 [P_n(x)]^2 dx - 2 \sum z^{m+n} \int_{-1}^1 P_n(x) P_m(x) dx$$

$$= \int_{-1}^1 \frac{1}{1 - 2xz + z^2} dx$$

$$\Rightarrow \sum z^{2n} \int_{-1}^1 [P_n(x)]^2 dx - 0 = \int_{-1}^1 \frac{1}{1 - 2xz + z^2} dx$$

$$\text{Let, } u = 1 - 2xz + z^2$$

$$\Rightarrow \frac{du}{dx} = -2z$$

$$\Rightarrow dx = \frac{-du}{2z}$$

$$\sum z^{2n} \int_{-1}^1 [P_n(x)]^2 dx = \frac{-1}{2z} \int_{-1}^1 \frac{1}{u} du$$

$$\Rightarrow \sum z^{2n} \int_{-1}^1 [P_n(x)]^2 dx = \frac{-1}{2z} [\log u]_{-1}^1$$

$$\Rightarrow \sum z^{2n} \int_{-1}^1 [P_n(x)]^2 dx = \frac{-1}{2z} [\log(1 - 2xz + z^2)]_1^1$$

$$\Rightarrow \sum z^{2n} \int_{-1}^1 [P_n(x)]^2 dx = \frac{-1}{2z} [\log(1 - 2z + z^2) - \log(1 + 2z + z^2)]$$

$$= \frac{-1}{2z} \log \frac{(1-z)^2}{(1+z)^2}$$

$$= \frac{1}{z} \log \frac{1+z}{1-z}$$

$$= \frac{1}{z} [\log(1+z) - \log(1-z)]$$

$$= \frac{1}{z} \left[\left(z - \frac{z^2}{2} + \frac{z^3}{3} - \dots \right) - \left(-z - \frac{z^2}{2} - \frac{z^3}{3} - \dots \right) \right]$$

$$= \frac{1}{z} \left[z - \cancel{\frac{z^2}{2}} + \frac{z^3}{3} - \dots + z + \cancel{\frac{z^2}{2}} + \frac{z^3}{3} + \dots \right]$$

$$= \frac{1}{z} \left[2z + \frac{2z^3}{3} + \dots \right]$$

$$= 2 \left[1 + \frac{z^2}{3} + \frac{z^4}{5} + \dots + \frac{z^{2n}}{2n+1} \right]$$

$$= \sum_{n=0}^{\infty} \frac{2}{2n+1} \cdot z^{2n}$$

Equating both sides, we get,

$$\int_{-1}^1 [P_n(x)]^2 dx = \frac{2}{2n+1} \text{ proved.}$$

Questions

Q.1) Show that: $\int_{-1}^1 [P_3(x)]^2 dx = \frac{2}{7}$

Solⁿ:

We have,

$$\int_{-1}^1 [P_n(x)]^2 dx = \frac{2}{2n+1}$$

Putting $n=3$, we get,

$$\int_{-1}^1 [P_3(x)]^2 dx = \frac{2}{7} \quad \underline{\text{proved}}$$

Q.2) Show that: $\int_{-1}^1 x P_n(x) \cdot P_{n-1}(x) dx = \frac{2n}{4n^2-1}$

Solⁿ:

From Recurrence Relation I,

$$(n+1) P_{n+1}(x) = (2n+1)x P_n(x) - n P_{n-1}(x)$$

Replacing n with $(n-1)$,

$$n P_n = (2n-1)x P_{n-1} - (n-1) P_{n-2}$$

$$\Rightarrow x P_{n-1} = \frac{1}{2n-1} [n P_n + (n-1) P_{n-2}]$$

$$\Rightarrow \int_{-1}^1 P_n(x) [x P_{n-1}(x)] dx$$

$$= \int_{-1}^1 P_n(x) \left\{ \frac{1}{2n-1} [n P_n(x) + (n-1) P_{n-2}(x)] \right\} dx$$

$$= \frac{n}{2n-1} \int_{-1}^1 [P_n(x)]^2 dx + \frac{(n-1)}{(2n-1)} \int_{-1}^1 P_n(x) P_{n-2}(x) dx$$

$$= \frac{n}{2n-1} \left(\frac{2}{2n+1} \right) + 0$$

$$\Rightarrow \int_{-1}^1 x P_n(x) \cdot P_{n-1}(x) dx = \frac{2n}{4n^2-1} \quad \underline{\text{proved}}$$